Task 3: SigmaPol Data Warehouse - relational design

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1. Business Process

The data warehouse is designed for the car rental business process, together with the fleet servicing process, whose data is needed to calculate the idle time and the utilization rate of cars. These processes are described in the documents *Specification of business processes in SigmaPol Rental network* and *Requirements specification for Rent business process*.

Client's satisfaction with a rented car is taken from the post-rental survey proposed in the requirements specification (Survey Excel source).

The data warehouse uses a **snowflake schema** with one fact table, CAR_DAY. The dimensions CAR and EMPLOYEE are snowflaked with the dimension LOCATION.

2. Relational Database Schema / Tables Specification

Fact Table

CAR_DAY (Fact Table)

One tuple describes the state of one car of the fleet on one day: whether the car was rented, in the service or idle. The rent and the service starting on this day are also described in this tuple (the values for a rent or a service are stored only in the tuple of the day it starts, so they are not counted twice).

- **Id_Date (Numeric, FK Date):** Day described by the tuple.
- **Id_Car (Numeric, FK Car):** Car described by the tuple.
- **Id_Employee (Numeric, FK Employee):** Employee who handled the rent the car is in on this day (special tuple "not rented" if the car is not rented).
- **Id_Client (Numeric, FK Client):** Client who rents the car on this day (special tuple "not rented" if the car is not rented).
- **Id_IssueTime (Numeric, FK Time):** Issue time of the rent starting on this day (special tuple "no rent" if no rent starts on this day).
- **Id_Junk (Numeric, FK Junk):** Junk attributes.
- **RentNo (Numeric):** Number of the rent the car is in on this day (empty if the car is not rented).
- **ServiceNo (Numeric):** Number of the service the car is in on this day (empty if the car is not in the service).
- **RentedDay (Numeric):** 1 if the car is rented on this day, otherwise 0.

- **ServiceDay (Numeric):** 1 if the car is in the service on this day, otherwise 0.
- **AvailableDay (Numeric):** 1 if the car is available for rental on this day (not in the service), otherwise 0.
- **IdleDay (Numeric):** 1 if the car is available but not rented on this day, otherwise 0.
- **Mileage (Numeric):** Mileage of the car at the beginning of the day in km.
- **KmDriven (Numeric):** Number of kilometres driven on this day (km of the rent divided by its days).
- **RentStarted (Numeric):** 1 if a rent of the car is issued on this day, otherwise 0.
- **RentCost (Money):** Total cost of the rent issued on this day (0 if no rent is issued).
- **RentalDays (Numeric):** Number of days of the rent issued on this day (0 if no rent is issued).
- **DelayDays (Numeric):** Number of days the rent issued on this day was returned after the planned return date (0 if on time or no rent is issued).
- **ServiceStarted (Numeric):** 1 if a service of the car starts on this day, otherwise 0.
- **ServiceCost (Money):** Total cost of the service starting on this day (0 if no service starts).
- **KmSinceLastService (Numeric):** Number of kilometres driven since the previous service, for the service starting on this day (0 if no service starts).
- **SurveyAnswered (Numeric):** 1 if the client answered the post-rental survey for the rent issued on this day, otherwise 0.
- **CarConditionRating (Numeric):** Client's rating of the overall condition of the car (values from 1 to 5) for the rent issued on this day (0 if the survey was not answered).
- **IssuesReported (Numeric):** 1 if the client experienced mechanical issues during the rent issued on this day, otherwise 0.
- **LikelihoodToRentAgain (Numeric):** How likely the client is to rent this car model again (values from 1 to 5) for the rent issued on this day (0 if the survey was not answered).

Dimension Tables

CAR (Dimension Table)

One tuple describes one car of the fleet, assigned to one rental point, within one price category.

- **Id_Car (Numeric, PK):** Surrogate key.
- **Id_Location (Numeric, FK Location):** Rental point the car is assigned to.
- **LicencePlate (Varchar(10), BK):** Business key, licence plate of the car.
- **Brand (Varchar(20)):** Brand of the car.
- **Model (Varchar(20)):** Model of the car.
- **Type (Varchar(10)):** Type of the car. Allowed values: suv, van, combi, sedan, coupe, sport.
- **FuelType (Varchar(10)):** Fuel type of the car. Allowed values: petrol, diesel, LPG, electric, hybrid.
- **Color (Varchar(10)):** Color of the car.
- **ProductionYear (4 digits):** Year of the car production.
- **PriceCategory (Varchar(15)):** Price category of the car based on its daily price. Allowed values: economy, standard, premium, luxury.

- **IsCurrent (Boolean):** 1 if information is current, otherwise 0 (SCD2 implementation).

LOCATION (Dimension Table)

One tuple describes one rental point.

- **Id_Location (Numeric, PK):** Surrogate key.
- **LocationName (Varchar(40), BK):** Business key, name of the rental point.
- **City (Varchar(20)):** City where the rental point is located.
- **District (Varchar(20)):** District of the city where the rental point is located.
- **Street (Varchar(30)):** Street and number of the rental point.
- **Status (Varchar(10)):** Status of the rental point. Allowed values: open, closed.

EMPLOYEE (Dimension Table)

One tuple describes one employee working in the specified rental point.

- **Id_Employee (Numeric, PK):** Surrogate key.
- **Id_Location (Numeric, FK Location):** Rental point the employee works in.
- **EmployeeNo (Numeric, BK):** Business key, employee identifier.
- **Name (Varchar(20)):** Name of the employee.
- **Surname (Varchar(30)):** Surname of the employee.
- **NameAndSurname (Varchar(50)):** Name and surname.
- **IsCurrent (Boolean):** 1 if information is current, otherwise 0 (SCD2 implementation).

The table contains one special tuple “not rented” used for days when the car is not rented.

CLIENT (Dimension Table)

One tuple describes one client in the specified age category.

- **Id_Client (Numeric, PK):** Surrogate key.
- **PID (11 digits, BK):** Business key, Personal Identification Number PESEL.
- **Name (Varchar(20)):** Name of the client.
- **Surname (Varchar(30)):** Surname of the client.
- **NameAndSurname (Varchar(50)):** Name and surname.
- **Gender (Varchar(10)):** Gender of the client. Allowed values: male, female.
- **DateOfBirth (Date):** Date of birth.
- **AgeCategory (Varchar(10)):** Age category. Allowed values: 18-24, 25-34, 35-44, 45-54, 55-64, 65+.
- **IsCurrent (Boolean):** 1 if information is current, otherwise 0 (SCD2 implementation).

The table contains one special tuple “not rented” used for days when the car is not rented.

DATE (Dimension Table)

One tuple describes one day.

- **Id_Date (Numeric, PK):** Date identifier.
- **Date (Date):** Calendar date.
- **Year (4 digits):** Year.
- **Quarter (Numeric):** Quarter of the year (values from 1 to 4).
- **Month (Varchar(10)):** Month name (January, February, …, December).
- **MonthNo (Numeric):** Month's numeric value.
- **DayOfWeek (Varchar(10)):** Day of week (Monday, Tuesday, …, Sunday).
- **DayOfWeekNo (Numeric):** Weekday's numeric value.
- **WorkingDay (Varchar(15)):** Allowed values: day off, working day.
- **Season (Varchar(10)):** Season of the year. Allowed values: winter, spring, summer, autumn.
- **Vacation (Varchar(20)):** Vacation time characteristics. Allowed values: non-holiday, winter holiday, summer holiday.
- **Holiday (Varchar(50)):** Type of public holiday. Allowed values: none, New Year, Epiphany, Easter, Easter Monday, Labour Day, Constitution Day, Pentecost, Corpus Christi, Assumption Day, All Saints' Day, Independence Day, Christmas Eve, Christmas, Second Day of Christmas.
- **BeforeHolidayDay (Varchar(62)):** Allowed values: none, tomorrow is New Year, tomorrow is Easter, etc.

TIME (Dimension Table)

One tuple describes one hour (independent of date).

- **Id_Time (Numeric, PK):** Time identifier.
- **Hour (Numeric):** Hour (values from 0 to 23).
- **TimeOfDay (Varchar(20)):** Time of day. Allowed values: between 0 and 8, between 9 and 12, between 13 and 15, between 16 and 20, between 21 and 23.

The table contains one special tuple "no rent" used for days when no rent is issued.

JUNK (Dimension Table)

The tuples correspond to all possible combinations of values for ReturnStatus, RentalLength, ServiceType, MileageCategory and CarAgeCategory (ServiceKind results from ServiceType).

- **Id_Junk (Numeric, PK):** Junk identifier.
- **ReturnStatus (Varchar(10)):** Return status of the rent the car is in. Allowed values: on time, late, not rented.
- **RentalLength (Varchar(20)):** Length of the rent the car is in. Allowed values: 1 to 3 days, 4 to 7 days, 8 to 14 days, more than 14 days, not rented.
- **ServiceType (Varchar(15)):** Type of the service the car is in. Allowed values: routine, accident, malfunction, none.
- **ServiceKind (Varchar(15)):** Kind of the service. Allowed values: planned (routine), unexpected (accident, malfunction), none.
- **MileageCategory (Varchar(25)):** Mileage of the car on the day. Allowed values: up to 50 000 km, 50 000 to 100 000 km, 100 000 to 200 000 km, over 200 000 km.

- **CarAgeCategory (Varchar(20)):** Age of the car on the day. Allowed values: up to 1 year, 1 to 3 years, 3 to 5 years, over 5 years.

Hierarchies

Besides the date hierarchy (Year → Quarter → Month → Date) and the time hierarchy (TimeOfDay → Hour), the data warehouse contains the following hierarchies:

- **Location hierarchy:** City → District → Rental point (LOCATION).
- **Car hierarchy:** Brand → Model → Car (CAR).
- **Organizational hierarchy:** Rental point (LOCATION) → Employee (EMPLOYEE).

3. Checking Data Availability in Data Sources (ETL Mapping)

The data sources are the **RentMaster** relational database (tables Rent, Car, Employee, Location and Service), the **CEO Excel** file (Sheet 1, information about clients) and the **Survey Excel** file (results of the post-rental survey proposed in the requirements specification, one row per rent: column A rent_id, column B car_condition_rating, column C experienced_issues, column D likelihood_to_rent_again, column E additional_comments).

CAR_DAY

The tuples are generated for every car and every day the car is in the fleet: from 1 January of car_year (or the first day of data) in Car table to the last day of data. “The rent covering the day” is the rent of the car from Rent table with date_issued (inclusive) ≤ day < date_returned_actual (exclusive), “the service covering the day” is the service of the car from Service table with start_date ≤ day ≤ end_date.

- **Id_Date:** Foreign key from dimension table. Generated day by day.
- **Id_Car:** Foreign key from dimension table. Based on licence_plate in Car table in RentMaster source and the version of the car (rental point, price category) valid on the day.
- **Id_Employee:** Foreign key from dimension table. Based on employee_id of the rent covering the day in Rent table in RentMaster source and the version of the employee valid on the day (“not rented” if there is no such rent).
- **Id_Client:** Foreign key from dimension table. Based on client_PESEL of the rent covering the day in Rent table in RentMaster source and the age category of the client on the day (“not rented” if there is no such rent).
- **Id_IssueTime:** Foreign key from dimension table. Based on the hour of date_issued of the rent issued on the day in Rent table in RentMaster source (“no rent” if there is no such rent).
- **Id_Junk:** Foreign key from dimension table. Based on date_returned_planned and date_returned_actual of the rent covering the day in Rent table, type of the service covering the day in Service table, Mileage of the tuple, and the age of the car calculated from car_year in Car table in RentMaster source and the day.
- **RentNo:** Taken from rent_id of the rent covering the day in Rent table in RentMaster source.
- **ServiceNo:** Taken from service_id of the service covering the day in Service table in RentMaster source.

- **RentedDay:** 1 if there is a rent covering the day in Rent table in RentMaster source, otherwise 0.
- **ServiceDay:** 1 if there is a service covering the day in Service table in RentMaster source, otherwise 0.
- **AvailableDay:** Calculated as 1 − ServiceDay.
- **IdleDay:** Calculated as AvailableDay − RentedDay.
- **Mileage:** Based on mileage_before and mileage_after in Rent table in RentMaster source: mileage_before of the rent covering the day, otherwise mileage_after of the last finished rent of the car (for days before the first rent: mileage_before of the first rent).
- **KmDriven:** Calculated for the rent covering the day as (mileage_after − mileage_before) divided by the number of days between date_issued and date_returned_actual in Rent table in RentMaster source.
- **RentStarted:** 1 if date_issued of a rent of the car in Rent table in RentMaster source is the day, otherwise 0.
- **RentCost:** Taken from cost column of the rent issued on the day in Rent table in RentMaster source.
- **RentalDays:** Calculated as the number of days between date_issued and date_returned_actual of the rent issued on the day in Rent table in RentMaster source.
- **DelayDays:** Calculated as the number of days between date_returned_planned and date_returned_actual of the rent issued on the day in Rent table in RentMaster source (0 if the car was returned on time).
- **ServiceStarted:** 1 if start_date of a service of the car in Service table in RentMaster source is the day, otherwise 0.
- **ServiceCost:** Taken from cost column of the service starting on the day in Service table in RentMaster source.
- **KmSinceLastService:** Calculated as the difference between mileage of the service starting on the day and mileage of the previous service of the same car in Service table in RentMaster source (for the first service: the difference to mileage_before of the first rent of the car in Rent table).
- **SurveyAnswered:** 1 if column A (rent_id) of Survey Excel source contains rent_id of the rent issued on the day, otherwise 0.
- **CarConditionRating:** Taken from column B (car_condition_rating) of Survey Excel source for the rent issued on the day.
- **IssuesReported:** Taken from column C (experienced_issues) of Survey Excel source for the rent issued on the day (yes → 1, no → 0).
- **LikelihoodToRentAgain:** Taken from column D (likelihood_to_rent_again) of Survey Excel source for the rent issued on the day.

CAR SCD 2

- **Id_Car:** Surrogate key generated by database.
- **Id_Location:** Foreign key from dimension table based on location_id in Car table in RentMaster source.
- **LicencePlate:** Business key taken from licence_plate in Car table in RentMaster source.
- **Brand:** Taken from Brand column in Car table in RentMaster source.

- **Model:** Taken from Model column in Car table in RentMaster source.
- **Type:** Taken from type column in Car table in RentMaster source.
- **FuelType:** Taken from fuel_type column in Car table in RentMaster source.
- **Color:** Taken from Color column in Car table in RentMaster source.
- **ProductionYear:** Taken from car_year column in Car table in RentMaster source.
- **PriceCategory:** Derived from Price_per_day in Car table in RentMaster source:
  - < 200: economy
  - < 300: standard
  - < 400: premium
  - = 400: luxury
- **IsCurrent:** '1' if information is current, otherwise '0'. A new tuple is created when the rental point (location_id) or the price category of the car changes.

LOCATION

- **Id_Location:** Surrogate key generated by database.
- **LocationName:** Business key taken from Name column in Location table in RentMaster source.
- **City:** Taken from City column in Location table in RentMaster source.
- **District:** Derived from Name column in Location table in RentMaster source (the part after the city name, e.g. "SigmaPol Kraków Nowa Huta" → Nowa Huta).
- **Street:** Taken from Street column in Location table in RentMaster source (overwritten when the rental point moves to a new address).
- **Status:** 'open' if the rental point exists in Location table in RentMaster source, 'closed' if it was removed from the table.

EMPLOYEE SCD 2

- **Id_Employee:** Surrogate key generated by database.
- **Id_Location:** Foreign key from dimension table based on location_id in Employee table in RentMaster source.
- **EmployeeNo:** Business key taken from employee_id in Employee table in RentMaster source.
- **Name:** Taken from Name column in Employee table in RentMaster source.
- **Surname:** Taken from Surname column in Employee table in RentMaster source.
- **NameAndSurname:** Taken from Name and Surname columns in Employee table in RentMaster source.
- **IsCurrent:** '1' if information is current, otherwise '0'. A new tuple is created when the employee moves to another rental point (location_id changes).

CLIENT SCD 2

- **Id_Client:** Surrogate key generated by database.
- **PID:** Business key taken from column C (PESEL) of Sheet 1 in CEO Excel source.

- **Name:** Taken from column A of Sheet 1 in CEO Excel source.
- **Surname:** Taken from column B of Sheet 1 in CEO Excel source (overwritten when changed).
- **NameAndSurname:** Taken from columns A and B of Sheet 1 in CEO Excel source.
- **Gender:** Taken from column E of Sheet 1 in CEO Excel source.
- **DateOfBirth:** Calculated from column C (PESEL) of Sheet 1 in CEO Excel source (first 6 digits).
- **AgeCategory:** Calculated from DateOfBirth and the day.
- **IsCurrent:** '1' if information is current, otherwise '0'. A new tuple is created when the client moves to the next age category.

DATE, TIME, JUNK

- **DATE:** Generated tuple by tuple based on calendar prior to ETL. Holiday, BeforeHolidayDay and Vacation are based on the public calendar of Polish public holidays and school holidays (e.g. Google Calendar).
- **TIME:** Generated tuple by tuple based on clock prior to ETL.
- **JUNK:** Generated with all combinations of ReturnStatus, RentalLength, ServiceType, MileageCategory and CarAgeCategory prior to ETL (ServiceKind: 'planned' for routine, 'unexpected' for accident and malfunction, 'none' if there is no service).

Data not loaded into the data warehouse

- Additional comments (column E of Survey Excel source) are free text and are not needed for any analytical question.
- Client's phone number and driver's licence number (columns F and G of Sheet 1 in CEO Excel source) are available, but they are not needed for any analytical question, so they are not loaded into the data warehouse.